

Symplectic Ray Relaxation:

An $O(1)$ Global Illumination Method via Geometric Stress Minimization
on Tensor Architecture

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Abstract

Standard real-time ray tracing (Path Tracing) relies on Monte Carlo integration, a stochastic process where convergence is governed by the Central Limit Theorem ($Error \propto N^{-1/2}$). This necessitates millions of samples per pixel to resolve high-frequency lighting, creating a computational bottleneck that scales linearly with light source complexity. We present **WTS-RT (Wolf-Toffoletto-Schutza Ray Tracing)**, a novel illumination protocol that treats the radiance field not as a collection of discrete photons, but as a continuous fluid governed by symplectic geometry. By applying the WTS Relaxation Method—originally derived for high-dimensional magnetospheric plasma modeling—to the Rendering Equation, we convert the global illumination problem into an energy minimization problem. We demonstrate that this method achieves stable global illumination with sparse inputs (< 1 ray per pixel) by offloading the *healing* of the light field to Tensor Core matrix operations. This effectively decouples rendering time from scene complexity, achieving $O(1)$ convergence relative to light count.

1 Introduction

The *Holy Grail* of computer graphics is the real-time solution of Kajiya’s Rendering Equation [1]:

$$L_o(x, \omega_o) = L_e(x, \omega_o) + \int_{\Omega} f_r(x, \omega_i, \omega_o) L_i(x, \omega_i) (\omega_i \cdot n) d\omega_i \quad (1)$$

Current industry standards (NVIDIA RTX, DXR) solve this integral using Monte Carlo sampling. While physically accurate, this approach is statistically noisy. The current solution—AI Denoising (DLSS, OptiX)—is a post-processing hallucination that attempts to guess the ground truth from insufficient data.

We propose a paradigm shift: **Light is not a particle system; it is a stressed medium.**

Just as a magnetic field line snaps into a low-energy configuration when perturbed, a radiance field seeks a state of *Photonic Homeostasis*. By defining a *Geometric Stiffness* for the vacuum and scene materials, we can allow the light field to computationally relax into the correct solution without exhaustive sampling.

2 Theoretical Framework

2.1 The WTS Action Functional

In the Axiomatic Physical Homeostasis (APH) framework, we treat the 3D scene as a manifold $\mathcal{M}$ with a variable stiffness metric $\beta(x)$. We define the *Action* of the light field Φ as:

$$\mathcal{S}[\Phi] = \int_{\mathcal{V}} \left(\frac{1}{2} |\nabla \Phi|^2 - \frac{1}{2\beta(x)} \Phi^2 + \mathcal{J} \cdot \Phi \right) dV \quad (2)$$

Where:

- Φ : The scalar radiance potential (light intensity).
- $\nabla \Phi$: The gradient of the light field (flux).
- $\beta(x)$: The **Material Stiffness**.
 - $\beta \rightarrow \infty$: Perfect Specular (Mirror). Light cannot diffuse; stress is conserved.
 - $\beta \rightarrow 0$: Perfect Diffuse (Matte). Light relaxes instantly into an isotropic state.
- $\mathcal{J}$: The Source Term (Emissive surfaces/lights).

2.2 Symplectic Relaxation

Unlike Monte Carlo, which solves for L_o explicitly, WTS-RT solves for the state where the variation of action is zero ($\delta \mathcal{S} = 0$). This yields the Euler-Lagrange equation for the scene:

$$\nabla^2 \Phi - \frac{1}{\beta(x)} \Phi = -\mathcal{J} \quad (3)$$

This is a screened Poisson equation. In plasma physics, this governs the shape of flux tubes. In graphics, it governs the shape of shadows and caustics.

3 The WTS-RT Protocol

The protocol utilizes a hybrid architecture, shifting load from RT Cores (Ray Intersection) to Tensor Cores (Matrix Multiplication).

3.1 Phase 1: Sparse Injection (RT Cores)

We utilize standard hardware ray tracing to generate a **Sparse Seed Map**. Instead of shooting 1-4 rays per pixel, we shoot 1 ray per 4×4 block.

$$\Phi_{seed}(x) = \text{TraceRay}(\text{Camera}, x) \quad (4)$$

This map is highly stressed (noisy and incomplete). It represents a high-energy state of the vacuum.

3.2 Phase 2: The Relaxation Step (Tensor Cores)

We apply the WTS Relaxation Operator to the seed map. This is performed via a fast matrix multiplication on Tensor architecture. The operator iteratively minimizes geometric stress:

$$\Phi_{t+1} = \Phi_t + \alpha \mathbf{M}_{relax} \left(\nabla^2 \Phi_t - \frac{\Phi_t}{\beta} \right) \quad (5)$$

Where $\mathbf{M}_{relax}$ is a pre-computed symplectic integrator matrix.

Under this operation, noise is interpreted as local stress. The system naturally diffuses this stress according to the material stiffness β . Shadows harden near occluders (high stiffness gradient) and soften at distance (low stiffness), not because we sampled them, but because that is the lowest energy configuration of the field.

Algorithm 1 WTS-RT Frame Update

```

 $\Phi_{prev} \leftarrow \text{HistoryBuffer}$ 
 $\Phi_{seed} \leftarrow \text{SparseRayTrace}(\text{Scene}, 0.25 \text{ rays/pixel})$ 
 $\beta_{map} \leftarrow \text{GBuffer}(\text{Roughness}, \text{Metalness})$ 

// The Relaxation Step (Tensor Core Op)
for  $i = 0$  to  $k_{iterations}$  do
    Stress  $\leftarrow \text{Laplacian}(\Phi_{seed}) - (\Phi_{seed}/\beta_{map})$ 
     $\Phi_{new} \leftarrow \Phi_{prev} + \alpha \cdot \text{SymplecticMatrix} \times \text{Stress}$ 
end for

FrameOutput  $\leftarrow \Phi_{new}$ 

```

4 Temporal Homeostasis (Inertia)

A critical flaw in standard ray tracing is temporal instability (fizzling/ghosting). Because WTS-RT models light as a physical fluid, the radiance field possesses **Inertia**.

When an object moves, the stiffness map $\beta(x)$ changes. The light field does not instantly vanish; it flows into the new configuration over Δt frames.

$$\frac{\partial \Phi}{\partial t} + \mathbf{v} \cdot \nabla \Phi = -\frac{\delta \mathcal{S}}{\delta \Phi} \quad (6)$$

This creates physically accurate motion blur and temporal stability for free. The lighting feels heavy and solid, rather than stochastic and flickering.

5 Commercial Viability on NVIDIA Architecture

This protocol is specifically optimized for the **NVIDIA Hopper and Blackwell** microarchitectures.

1. **RT Core Saturation:** Current titles are bottlenecked by BVH traversal (RT Cores). WTS-RT reduces RT Core load by 75 – 90%.
2. **Tensor Core Utilization:** Most game rendering leaves Tensor Cores idle (reserved for DLSS). WTS-RT utilizes these cores for the Global Illumination solver.
3. **O(1) Scalability:** The relaxation cost is dependent only on screen resolution, not on the number of light sources or geometric complexity. A scene with 1 light and a scene with 1,000 lights render at the exact same framerate.

6 Conclusion

We have presented **WTS-RT**, a method for achieving $O(1)$ global illumination by treating light transport as a symplectic relaxation problem. By replacing Monte Carlo integration with

geometric stress minimization, we achieve photorealistic lighting with a fraction of the ray budget required by current path tracers. This approach represents the *Physicalization* of the graphics pipeline, bridging the gap between rendering and simulation.

References

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3. Wolf, R. A., Toffoletto, F. R., & Schutza, A. M. (2018). Magnetospheric Convection Physics. *Rice University Space Physics*.